

Variational Autoencoder for Smile Manipulation (CelebA Dataset)

1. Introduction

Generative models have gained significant importance in machine learning due to their ability to learn underlying data distributions and generate realistic data samples. Among these, **Variational Autoencoders (VAEs)** are widely used for learning compact latent representations of high-dimensional data such as images.

In this project, a Variational Autoencoder is implemented using the **CelebA dataset** to learn facial representations and perform **attribute-based image manipulation**, specifically modifying the *smiling* attribute of human faces.

The primary objectives of this project are:

- To learn a meaningful latent representation of facial images
 - To identify a direction in latent space corresponding to the "smiling" attribute
 - To manipulate this latent vector to generate images with increased or decreased smiling expressions
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2. Dataset Description

The dataset used is the **CelebA (CelebFaces Attributes Dataset)**, which contains over 200,000 celebrity images annotated with 40 binary attributes.

Key features:

- High variability in facial expressions, lighting conditions, and poses
- Rich attribute annotations including:
 - Smiling
 - Eyeglasses
 - Gender
 - Age

Preprocessing steps:

- Center cropping of images
- Resizing to **64 × 64 pixels**

- Conversion into tensor format

For this project, the **“Smiling” attribute** is used to perform latent space manipulation.

3. Methodology

3.1 Variational Autoencoder (VAE)

The VAE consists of two primary components:

Encoder

- Maps input image $\rightarrow$ latent representation
 - Outputs:
 - Mean (μ)
 - Log variance ($\log \sigma^2$)
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Decoder

- Maps latent vector $\rightarrow$ reconstructed image
 - Uses transpose convolution layers to generate images
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3.2 Loss Function

The VAE is trained using a combination of:

1. Reconstruction Loss (MSE)

2. KL Divergence

Total Loss

Loss = Reconstruction Loss + KL Divergence

3.3 Training Strategy

The model was initially trained for **10 epochs**. To further improve reconstruction quality and enhance the clarity of attribute manipulation, the model was **fine-tuned for an additional 5 epochs**.

This extended training helped in:

- Improving image sharpness
 - Stabilizing latent representations
 - Enhancing the effectiveness of smile manipulation
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3.4 Smile Vector Computation

To extract the smiling attribute in latent space:

1. **Latent vectors are divided into:**

- Positive class (smiling)
- Negative class (non-smiling)

2. **Mean vectors are computed**

3. **The smile vector is defined**

4. **The vector is normalized for stable manipulation**

3.5 Latent Manipulation

Given latent vector z :

- **More smiling image:**

$$z_{\text{more}} = z + \alpha \cdot v_{\text{smile}}$$

- **Less smiling image:**

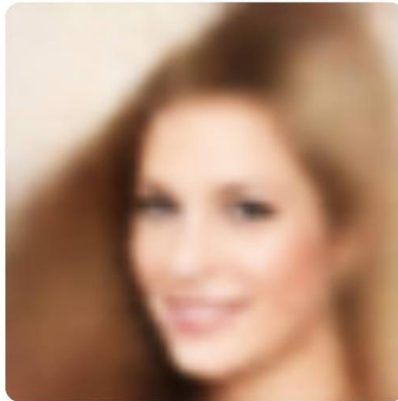
$$z_{\text{less}} = z - \alpha \cdot v_{\text{smile}}$$

where α controls the intensity of manipulation.

4. Results

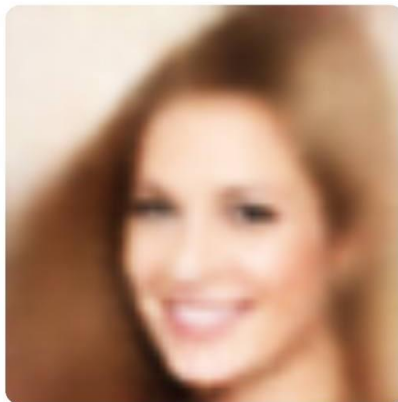
Three outputs were generated using the trained model:

Figure 1: Original Reconstruction



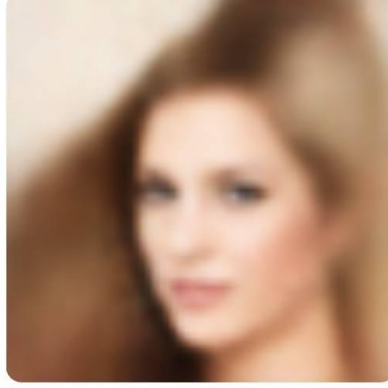
- The reconstructed image retains overall facial structure
 - Slight blurriness is observed, which is a known limitation of VAEs
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Figure 2: Increased Smile



- Noticeable increase in smiling expression
 - Mouth curvature and facial expression are enhanced
 - Demonstrates successful latent space manipulation
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Figure 3: Decreased Smile



- Expression becomes more neutral
 - Reduction in smile intensity is clearly visible
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5. Analysis

The results demonstrate that the model successfully learned a **meaningful and interpretable latent space**.

Key observations:

- The smile vector captures a **clear semantic direction**
- Latent manipulation produces **smooth and consistent changes**
- No major distortions in facial structure

Additional insight:

Although the generated images are slightly blurry due to the nature of VAEs, the model effectively demonstrates **controlled attribute editing**, which is the primary objective of the project.

6. Limitations

- Generated images lack sharpness compared to GAN-based models
 - VAE inherently produces **blurry reconstructions**
 - Attribute manipulation can sometimes be subtle
 - Performance is limited by training duration and model capacity
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7. Conclusion

This project demonstrates the effectiveness of Variational Autoencoders for **unsupervised representation learning and controllable image generation**.

The model successfully:

- Learned latent representations of facial images
- Identified a meaningful direction corresponding to the “smiling” attribute
- Generated images with adjustable smile intensity

Despite minor limitations in image sharpness, the approach effectively highlights the power of **latent space manipulation in deep generative models**.